Image-to-Image Translation for Clutter and Subsurface Discrimination in Radar Sounder Data

Samuele Trainotti

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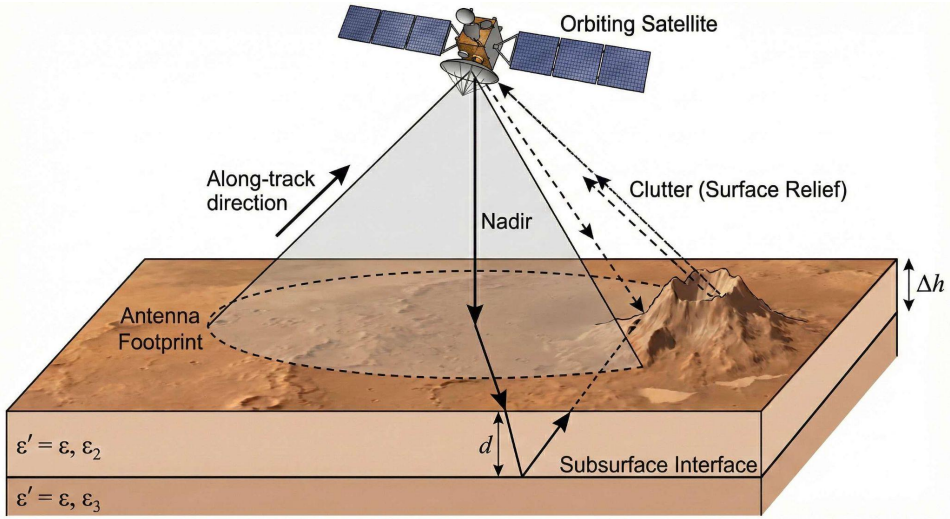
Supervisor: Bruzzone Lorenzo
Co-Supervisor: Dal Corso Jordy

Planetary Exploration & Data Volume

- **Orbital Radar Sounders**: Acquire terabytes of subsurface data to locate Martian ice and geological structures
- **The Data Bottleneck**: The data volume (tens of TBs) vastly exceeds manual interpretation limits
- **Goal**: Develop automated, data-driven analysis pipelines



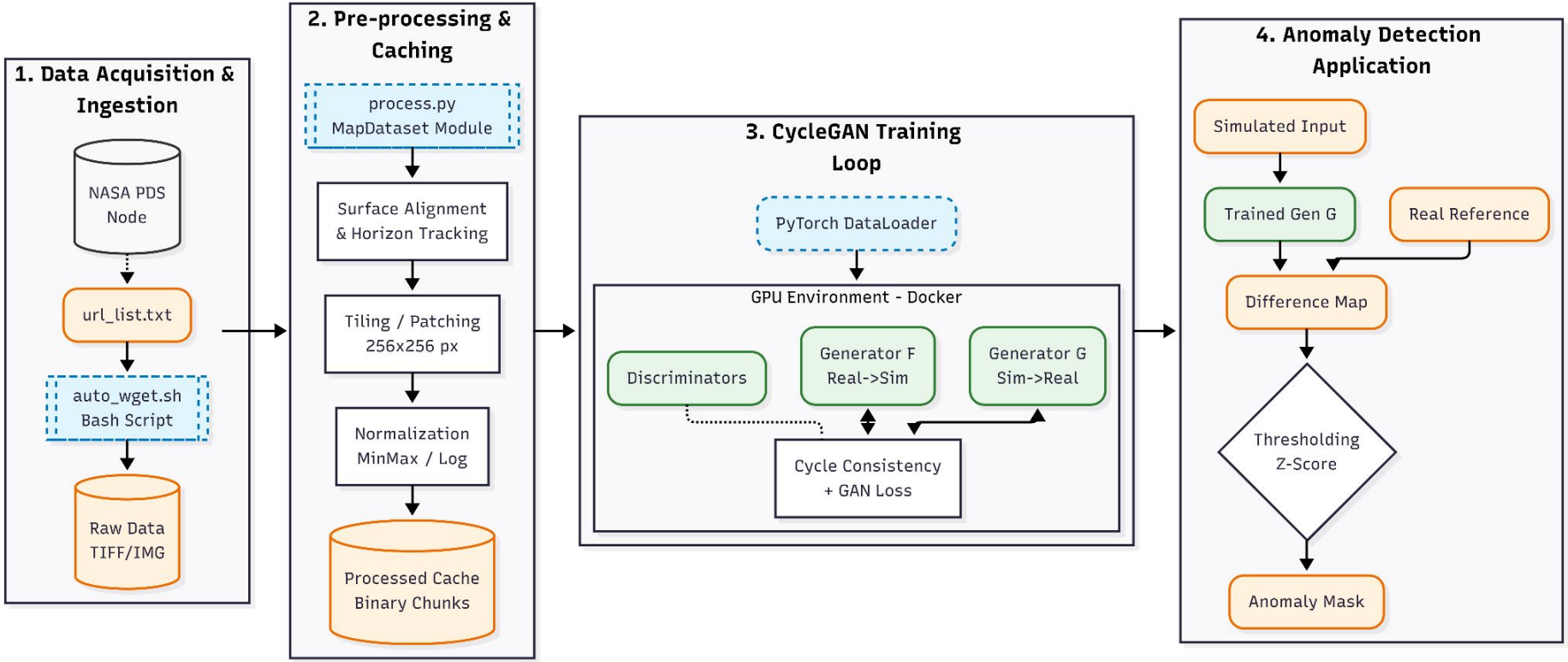
The Challenge of Surface Clutter

- **Surface Clutter**: Off-nadir topographical reflections mimic true underground structures
- **The Domain Gap**: Simulators predict clutter but lack noise, speckle, and attenuation
- **The Machine Learning Barrier**: No labeled ground truth on Mars for supervised learning

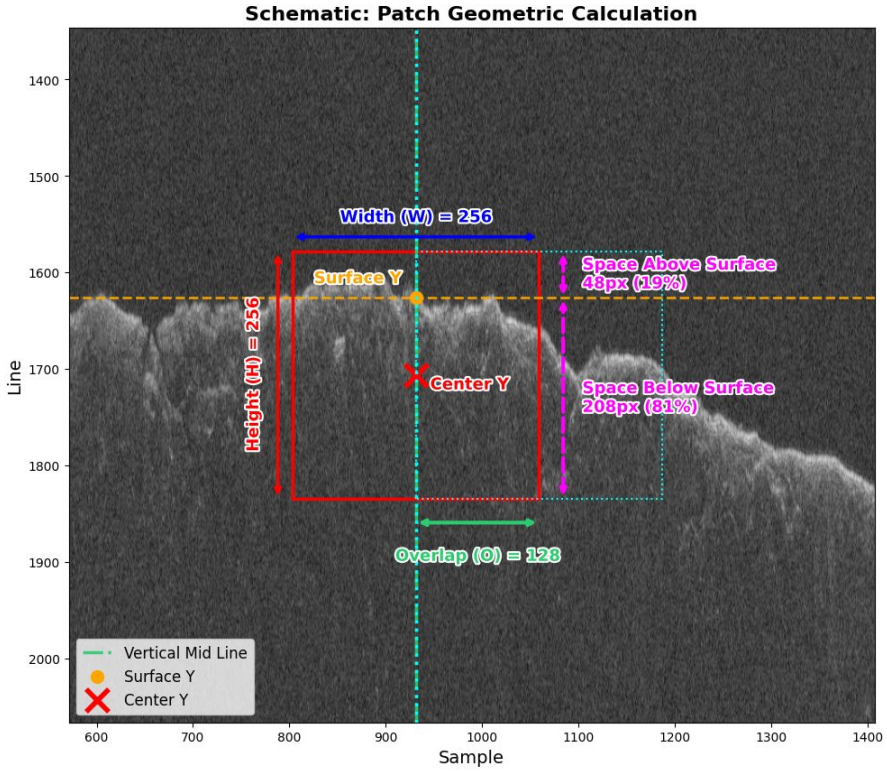
Unsupervised Anomaly Detection via Sim-to-Real Translation

- **Approach**: Sim-to-Real image translation
- **Hypothesis**: The network learns the "normal" clutter background. Unmodeled subsurface interfaces emerge as anomalies in the residual map

Unsupervised Anomaly Detection via Sim-to-Real Translation



Scalable Data Engineering: Dynamic Patching

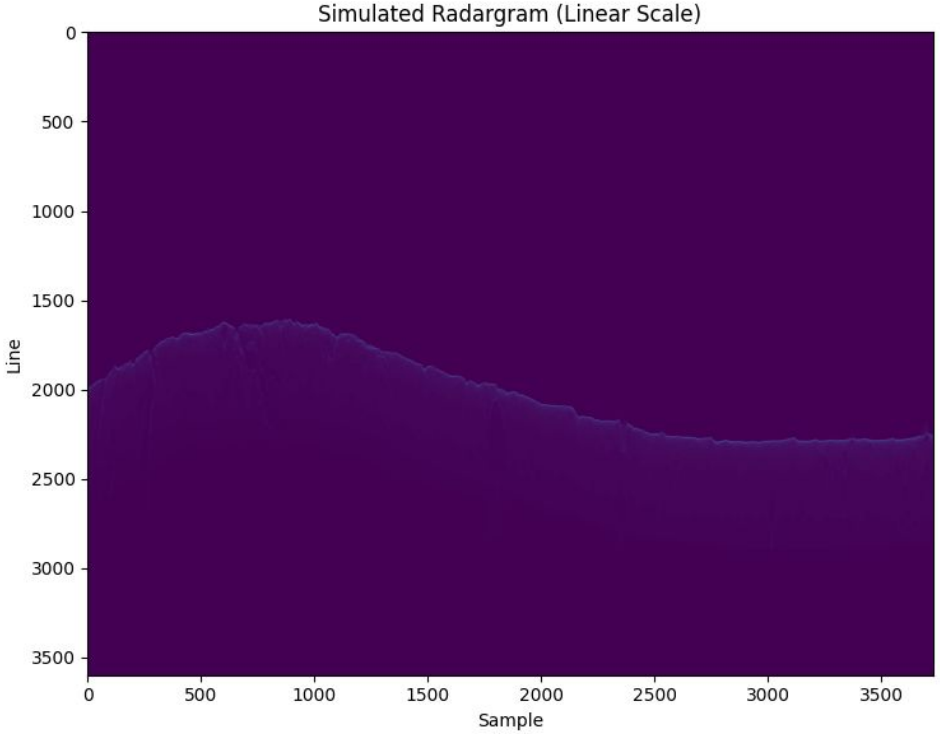


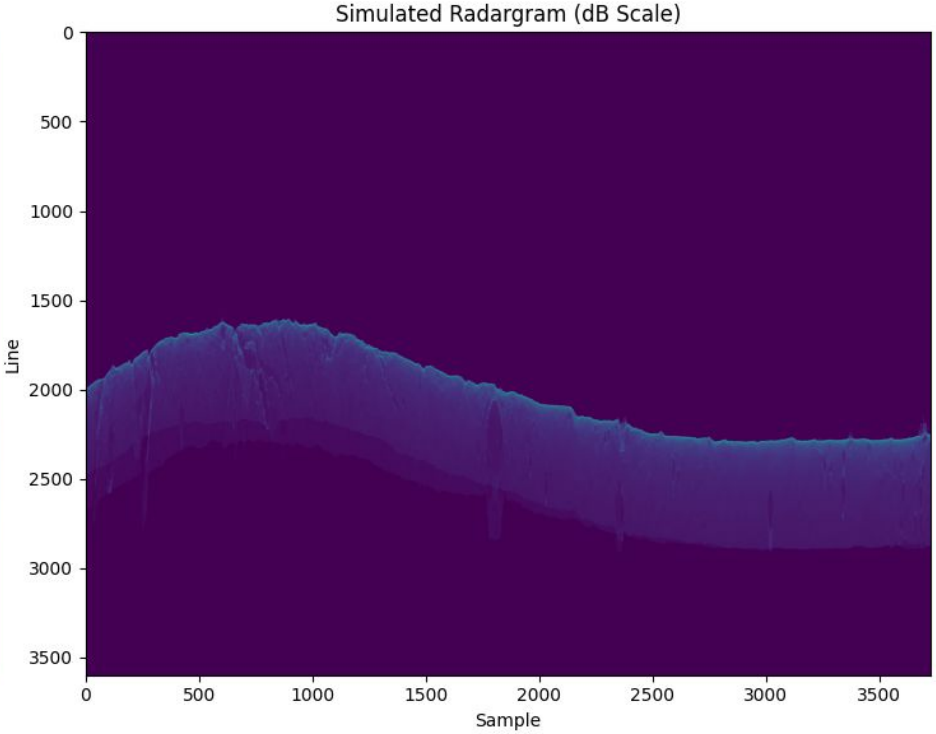
- **Surface-Aware Extraction**: Custom sliding window algorithm that dynamically tracks the Martian topography
- **Data Optimization**: Discards uninformative atmospheric regions, focusing strictly on the subsurface
- **Fast Inference Tracking**: Enables rapid processing of 128x128 patches (~4.2 ms per patch on RTX 3090 GPU)

Scalable Data Engineering: Normalization

- **Dynamic Range Mitigation**: Logarithmic (dB) scaling to compress the energy difference between surface and subsurface
- **Min-Max Scaling**: Applied 1st and 99th percentile clipping to suppress extreme reflections that destabilize GAN gradients
- **Reproducibility**: Fully Dockerized environment & Optuna hyperparameter tuning

Scalable Data Engineering: Normalization





Bridging the Gap: Unpaired Image Translation

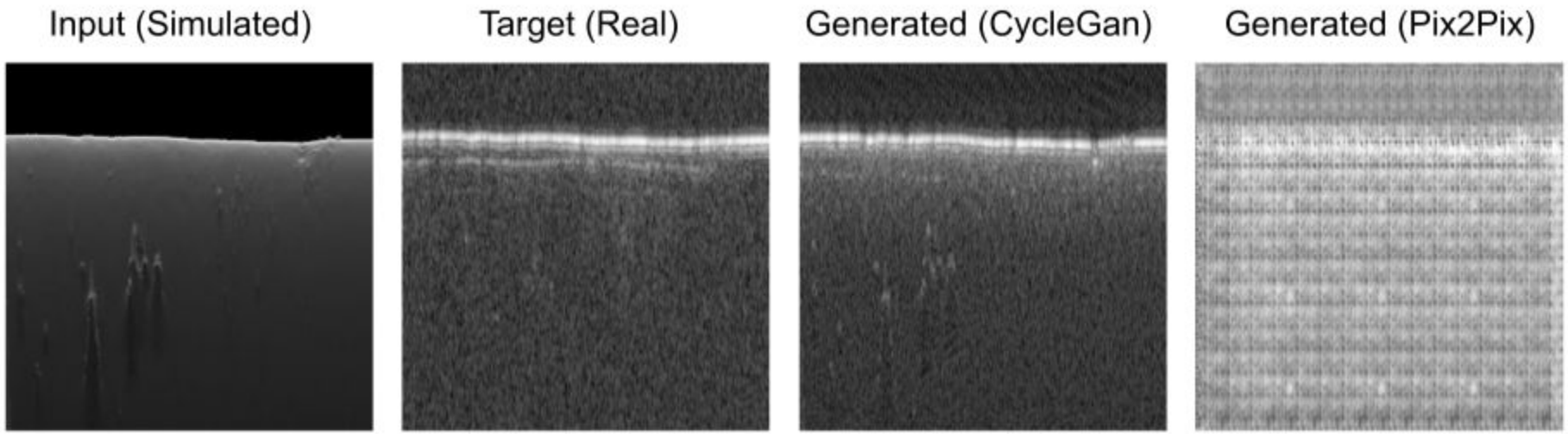
- **Unpaired Translation**: CycleGAN selected over Pix2Pix to relax spatial constraints and overcome unavoidable misalignments between simulator and real data
- **Cycle-Consistency**: Preserves input simulation structure while injecting realistic speckle
- **Generator (ResNet)**: Employs Reflection Padding instead of zero-padding to eliminate grid artifacts (edge effects) during patch reconstruction
- **Discriminator (PatchGAN)**: Enforces high-frequency texture realism to synthesize accurate speckle noise

Validation Framework: Synthetic Anomaly Injection

- **The Challenge**: No labeled ground truth exists for Martian subsurface anomalies (supervised validation is impossible)
- **The Solution**: Physics-informed Synthetic Anomaly Injection into real radargrams
- **Physical Constraints**: Injected targets follow depth-dependent attenuation and speckle interference laws
- **Advantage**: Provides exact binary masks to compute rigorous ROC and Precision-Recall metrics

Sim-to-Real Generative Quality

- CycleGAN successfully synthesizes realistic speckle and preserves topography
- Pix2Pix suffers from structural distortions due to imperfectly aligned data



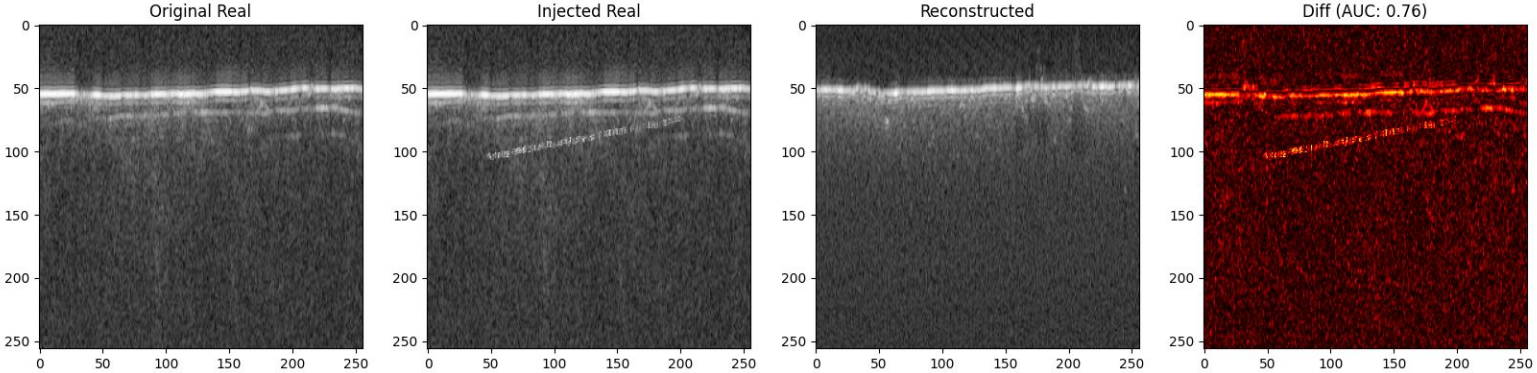
Quantitative Generative Metrics

- **Fréchet Inception Distance (FID)**: 1.56 (CycleGAN) vs 11.03 (Pix2Pix). Confirms high statistical realism
- **LPIPS**: 0.104. Validates excellent structural integrity during domain transfer

Model	PSNR [dB] ↑	SSIM ↑	LPIPS ↓	FID ↓
Pix2Pix	19.0	0.234	0.254	11.03
CycleGAN	**23.0**	**0.291**	**0.104**	**1.56**

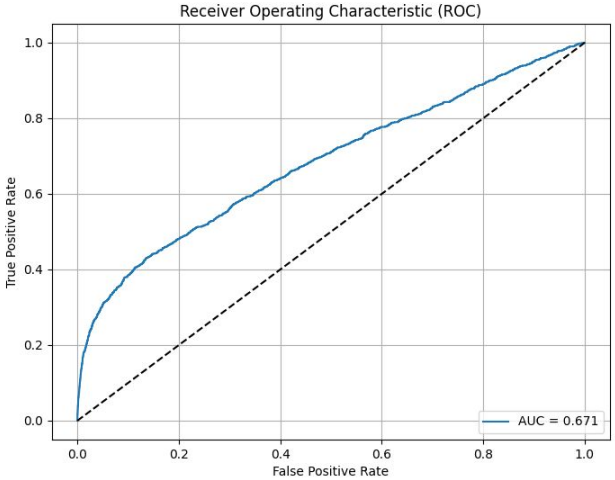
Clutter vs. Subsurface Discrimination

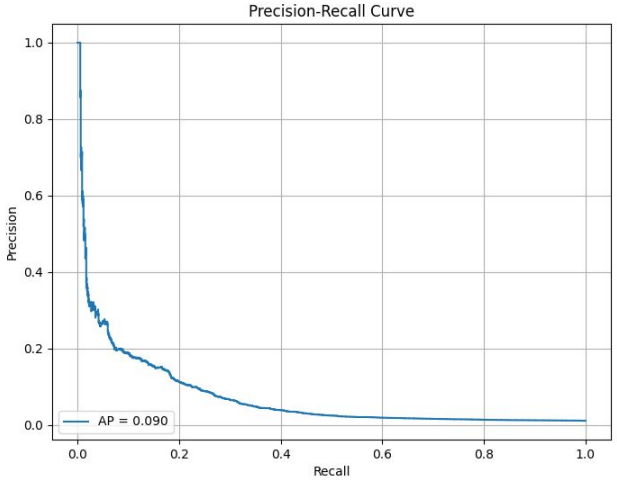
- **Real**: Contains clutter + injected anomaly
- **Generated**: Reconstructs only surface clutter
- **Residual Map**: Successfully isolates the unmodeled subsurface anomaly



Detection Performance

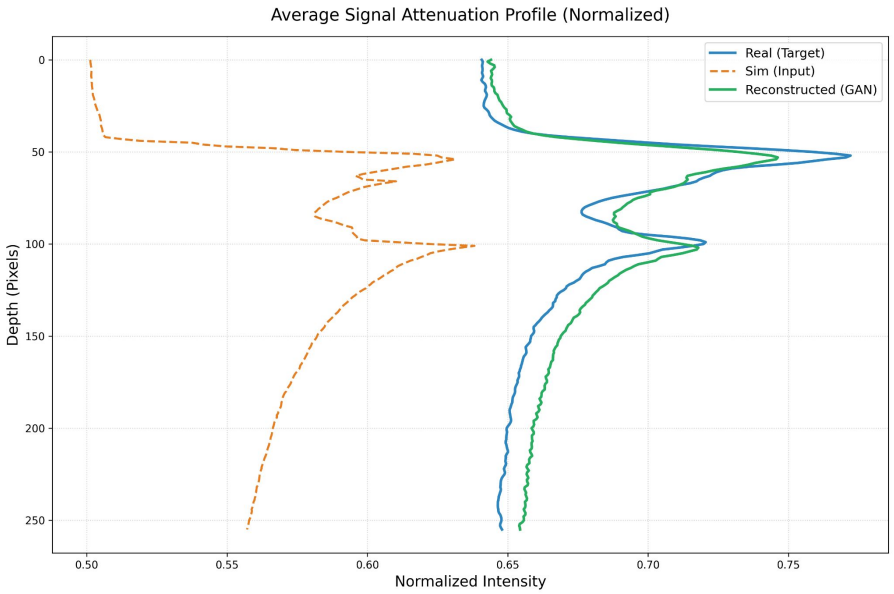
- Evaluated on 60 patches with injected anomalies
- **ROC AUC = 0.671**: Unsupervised discriminative capability
- **mAP = 0.089**: Highlights the inherent difficulty caused by severe speckle noise





Limitations: Vision Models vs. Physical Laws

- **Signal Attenuation**: GANs replicate local textures but struggle to explicitly learn global exponential energy decay
- **Engineering Mitigation**: Independent Z-score normalization applied before residual computation to remove vertical bias



Conclusions

- Developed a scalable, memory-safe data engineering pipeline
- Successfully bridged the Sim-to-Real domain gap using CycleGAN.
- Validated an unsupervised anomaly detection framework via synthetic injection

Future Works

- Physics-Informed loss to explicitly enforce energy decay equations
- Adaptive thresholding for residual maps
- Validation on verified geological datasets from diverse Martian regions
- Extension to future planetary missions (e.g., ESA JUICE, NASA Europa Clipper) to analyze icy moons

Thank you for your attention

Questions?